

Groundnut rosette disease

Groundnut Rosette Virus (GRV)

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Yellowing (chlorosis) and distortion of the leaves (Scottish Crop Research Institute)	- Plant rosette resistant or tolerant varieties such as 'Adepa' or 'Konkoma' released by CSRI (Crops Research Institute). Contact CSRI to see if they are available. - In the absence of resistant varieties the following can be practiced: - Remove volunteer groundnut or leftover plants immediately after harvest and destroy in the field by burning or burying. - Intercrop groundnuts with cereals such as sorghum, millet or maize. - Practice crop rotation ever 1-2 years with such as sorghum, millet or maize. - Practice early planting and avoid wide spacing. Plants should form uniformly dense stands as aphids often land preferentially on widely spaced plants.	- Monitor young plants weekly for aphids vectors. Examine leaf surfaces after germination particularly growing tips - aphids prefer feeding on young and soft tissue. - Look for glossy black or dark brown bodied aphids measuring about 2 mm in length on the plants. - Monitor young plants from 30-45 days after sowing for symptoms of rosette - stunted growth, chlorosis and reduced leaf sizes and leaf distortion. - Start control as soon as symptoms of the virus are observed or the presence aphids are detected. - Yellow direct control should not be delayed if susceptible cultivars or varieties are planted.	- Rogue infected plants when observed to reduce source of infection and destroy by burning in the field. - Control aphids by spraying soap solution (1 tea spoonful of dishwashing soap in 4L of water) 2-3 times at 3-4 day intervals early or late in the day. - Spray neem seed water extract (5% concentration, w/v) to control aphids just after their appearance (second and third sprays must be applied at 14 days intervals). PHI: 24 hrs. - Position a number of sticky yellow traps or yellow sheets (A4 size) at different locations in the field to trap winged aphids.	- When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label. - Do not use chemicals with the same mode of action year after year as this can lead to resistance. - Always consult the most recent list of registered pesticides of MOFA, Ghana. - Application of chemical control may not be practiced by small scale farmers for economic reasons.	- WHO class II (Moderately hazardous). Apply 30 days after sowing and repeat 15 days after first application to control population of aphids. PHI: 24 hrs & REI 24 hrs - WHO class II (moderately hazardous). Apply 30 days after sowing and repeat 15 days after first application. PHI: 24 hrs & REI 24 hrs
 Stunting of an infected plant on the right compared to a healthy plant on the left (Scottish Crop Research Institute)				- Spray imidacloprid (Confidor 200SL, Confidor 200 OD, Bastion Extra 3G, Akape 20 SC) at 5-8ml/15L of water to control aphids and other insect vectors. Systemic neonicotinoid, IRAC group:4A - Spray lambda-cyhalothrin (Clear 2.5 EC, Conquer Super 2.5 EC, Ekuapa 2.5 EC) at 50ml/15L of water to control aphids. A quick-acting contact and stomach action pyrethroid, IRAC group: 3A	

Ghana

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